

Scrum Methodology

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Agile Recap

Agile manifesto values:

- **Individuals & interactions** over processes & tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

Agile methodologies common features

- **Iterative, incremental delivery** (short cycles)
- **Continuous customer feedback** and prioritization
- **Adaptive planning**; embrace changing requirements
- **Small, cross-functional, empowered teams**

Scrum

- Most popular **Agile framework**
- Simple structure to achieve goals and create value
- **Empirical process control**: transparency, inspection, adaptation
- **Lean thinking**: minimize waste and focus on the essentials
- **Timeboxed iterations** (sprints), typically 2–4 weeks

Scrum overview

In a nutshell, Scrum requires a **Scrum Master** to foster an **environment** where:

1. A **Product Owner** orders the work for a complex problem into a **Product Backlog**.
2. The **Scrum Team** turns a selection of the work into an **Increment of value** during a **Sprint**.
3. **The Scrum Team and its stakeholders** inspect the results and adjust for the **next Sprint**.
4. **Repeat**

Scrum pillars

- **Transparency**
 - Process and work must be visible so everyone can inspect them
- **Inspection**
 - Frequent checks on progress toward goals
- **Adaptation**
 - Adjust process and plans based on inspection results

Scrum values

- **Commitment**
 - to achieving its goals and to supporting each other.
- **Focus**
 - on the work of the Sprint to achieve the goals.
- **Openness**
 - about the work and the challenges.
- **Respect**
 - everyone and expect to be respected.
- **Courage**
 - to do the right thing and to work on tough problems.

Scrum team

- Roles
 - **Scrum Master** (1)
 - **Product Owner** (1)
 - **Developers** (3–9)
- **Cross-functional, self-organizing**, collaborative.
- Accountable for delivering **value each Sprint**.
- **Shared responsibility** for quality and delivery.

Developers

- Create a plan for the Sprint, the **Sprint Backlog**
- Instill quality by adhering to a **Definition of Done**
- Adapt their plan each day toward the **Sprint Goal**
- Hold each other accountable as **professionals**

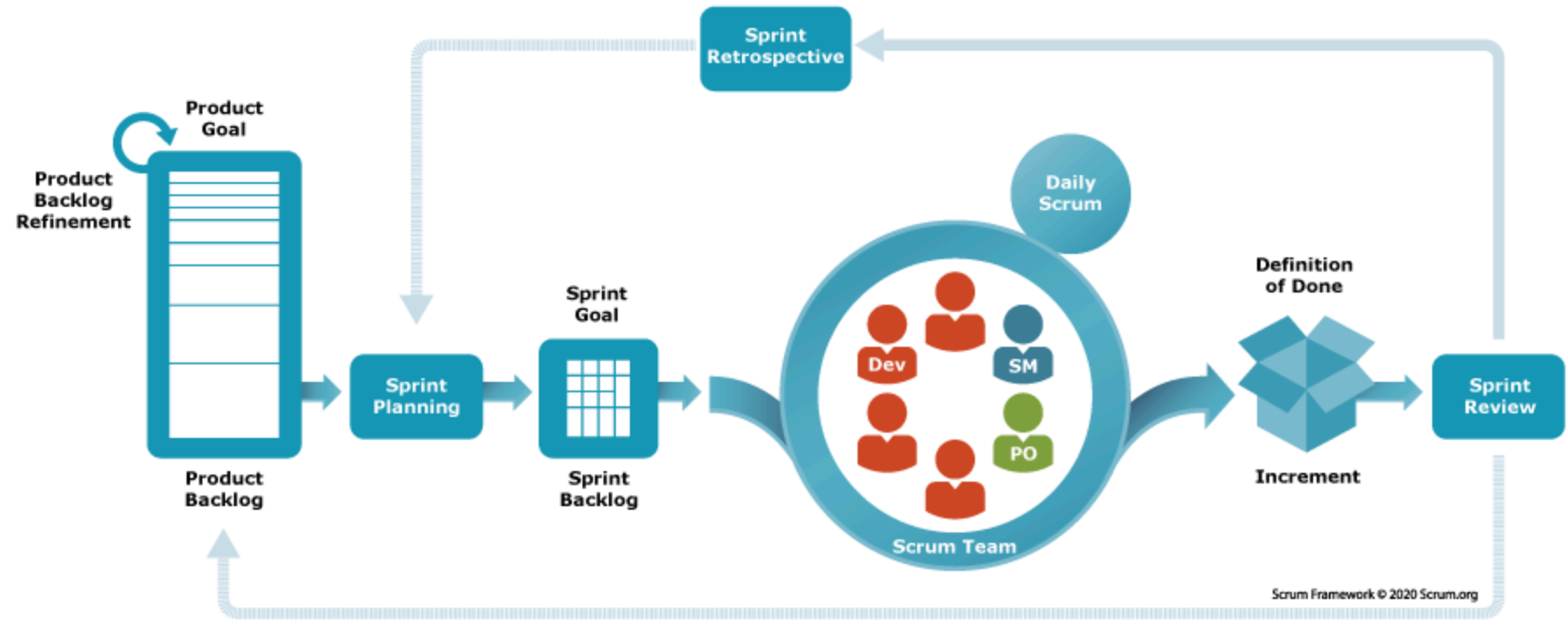
Product owner

- Defines and explicitly communicates the **Product Goal**
- Creates and clearly communicates **Product Backlog items**
- **Orders** Product Backlog items
- Ensures that the Product Backlog is **transparent, visible and understood**

Scrum master

- Accountable for establishing **Scrum and team effectiveness**
- **Coaches** self-management, cross-functionality, and adherence to DoD
- **Removes impediments** to team progress
- Ensures all Scrum **events** are **positive, productive, and timeboxed**
- Serves as a **leader** for the team, PO, and organization

Scrum events: The Sprint



Sprint

- **Back-to-back, fixed-length events** (≤ 1 month) where ideas turn into value.
- **Sprint Goal**: single objective for the Sprint.
- Enable **predictability** via periodic inspect & adapt
 - Shorter Sprints = more learning, less risk
- During a Sprint:
 - **No changes endangering Sprint Goal**
 - **Quality is maintained**
 - **The Backlog is refined**
 - **Scope is negotiated** with PO

Sprint Planning

- Initiates the Sprint; collaborative planning by entire Scrum Team
- **Topic 1 — Why?** Team defines Sprint Goal communicating value to stakeholders
- **Topic 2 — What?** Developers select Product Backlog items for the Sprint (informed by past performance, capacity, DoD)
- **Topic 3 — How?** Developers plan work and decompose items into ≤ 1 -day tasks
- Output: **Sprint Backlog** = Sprint Goal + selected items + delivery plan
- Timeboxed to max 8 hours for 1-month Sprint (shorter for shorter Sprints)

Daily Scrum

- **15 minutes a day** for Developers to inspect progress toward Sprint Goal
- Held **same time/place** each working day to reduce complexity
- Team defines structure, with focus on **Sprint Goal** and **actionable daily plan**
 - Typical structure:
 - What did I do **yesterday** to help meet the Sprint Goal?
 - What will I do **today** to help meet the Sprint Goal?
 - Do I see any **impediment** that prevents me or the team from meeting the Sprint Goal?
- Improves **communication**, identifies **impediments**, promotes **quick decisions**, eliminates extra meetings
- Not the only time to adapt: Developers meet throughout day for detailed **re-planning** as needed

Sprint Review

- **Inspect** Sprint outcome and determine **future adaptations**; discuss progress toward Product Goal
- Scrum Team **presents work results** to key stakeholders
- **Collaborative working session** (not just a presentation) to decide what to do next
- Product Backlog may be **adjusted** based on new opportunities/changes in environment
- **Timeboxed** to max 4 hours for 1-month Sprint (shorter for shorter Sprints)

Sprint Retrospective

- Purpose: plan ways to increase **quality and effectiveness**
- Team **inspects last Sprint**: individuals, interactions, processes, tools, DoD
- Discuss **what went well, what problems** occurred, how they were (or weren't) solved
- Identify **most impactful improvements** and address ASAP (may add to next Sprint Backlog)
- **Concludes the Sprint**; timeboxed to max 3 hours for 1-month Sprint (shorter for shorter Sprints)

Scrum artifacts

- Artifacts represent **work or value**; designed to maximize **transparency**
- Each artifact has a **commitment** to enhance transparency and focus:
 - **Product Backlog → Product Goal**
 - **Sprint Backlog → Sprint Goal**
 - **Increment → Definition of Done**
- Commitments reinforce **empiricism** and Scrum values

Product Backlog

- **Emergent, ordered list** of what's needed to improve the product
- **Single source of work** for the Scrum Team
- Items **refined** into smaller, precise items (description, order, size)
- **Developers responsible for sizing**; PO helps with trade-offs
- Items **"ready"** when they can be Done within one Sprint

Product Backlog Commitment: Product Goal

- Describes **future state** of the product; target for Scrum Team planning
- **Long-term objective**; must fulfill (or abandon) one before taking next
- Rest of Product Backlog emerges to define "**what**" fulfills the Product Goal
- Product = **vehicle to deliver value** (service, physical product, or abstract)

Sprint Backlog

- Composed of **Sprint Goal** (why) + selected **Product Backlog** items (what) + **delivery plan** (how)
- Plan **by and for Developers**; highly visible, **real-time picture** of Sprint work
- **Updated throughout Sprint** as more is learned
- Should have enough detail for **Daily Scrum inspection**

Sprint Backlog Commitment: **Sprint Goal**

- **Single objective** for the Sprint; creates **coherence and focus**
- **Commitment by Developers** with flexibility in exact work needed
- Created during **Sprint Planning**, added to Sprint Backlog
- Developers keep it in mind; **negotiate scope** with PO if needed (without affecting goal)

Increment

- **Concrete stepping stone** toward Product Goal
- **Additive** to all prior Increments; thoroughly verified to work together
- Must be **usable** to provide value
- **Multiple Increments** may be created within a Sprint
- May be **delivered before Sprint end**; Review is **not a gate** to releasing value
- Work **not meeting DoD** cannot be part of an Increment

Increment Commitment: Definition of Done

- **Formal description** of Increment's state when it meets quality measures
- Creates **transparency** via shared understanding of "completed" work
- Item meeting DoD → **Increment is born**
- Item not meeting DoD → **returns to Product Backlog**
- **Organization standard** (minimum) or Scrum Team creates one for product
- **All Developers must conform**; multiple teams share same DoD